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Brief paper

Hierarchical optimization-based online learning control for nonlinear systems with unknown disturbances☆



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ABSTRACT

A hierarchical optimization-based online learning control scheme is presented to exploit the learning control performance of neural networks (NNs) for nonlinear systems with unknown disturbances. First, leveraging the equivalent-input-disturbance (EID) concept and using only the system output, an NN-assisted control structure is devised. Then, a two-stage learning control framework is developed. Specifically, the first rough stage is for fast dynamic disturbance-rejection control with exponential convergence performance and then the second refined stage is to further refine the performance by online optimization. Further, the closed-loop performance with integrated disturbance-rejection performance is analyzed and a performance-oriented design algorithm is presented. Finally, superiorities of the developed scheme are shown by comparisons of rotary inverted pendulum experiments with other methods.

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1. Introduction

Nonlinearities and unknown disturbances, such as friction and payload changes of robotic manipulators (Chen et al., 2000), are ubiquitous in practice. They may induce undesired impacts on control systems. Hence, numerous control methods (Chen et al., 2016) are applicable.

Passive control methods, like sliding-mode control (Roy et al., 2020), utilize high gains to suppress the disturbance. However, high gains may excite system unmodeled dynamics.

Despite the differences between nonlinearities and disturbances and by treating their combined effect as an overall disturbance, active control methods, like the disturbance observer (DOB) (Song & Lewis, 2020), the extended-state observer (ESO) (Ma et al., 2021), and the equivalent-input-disturbance (EID) estimator (She et al., 2008), make use of an extra loop to estimate the overall disturbance and then feed the estimate back to an original control law to compensate for the nonlinearities and disturbances. Thereby, active control methods allow the transformation of the challenge of nonlinear system control into the problem of disturbance-rejection control.

The DOB employs an inverse of a nominal plant model combined with a filter to estimate matched disturbances in the frequency domain (Sariyildiz et al., 2020). The ESO treats matched disturbance as an extended state (Hui et al., 2021). The EID estimator is based on the concept of an EID (She et al., 2008), which is an artificial disturbance acting on the control input matrix but producing the same effect on the system output as actual disturbances do. Different from the DOB and ESO, the EID estimator is aimed to get an estimate of an EID. Thereby, the EID estimator is usually effective for both matched and mismatched nonlinearities and disturbances (Kürkçü et al., 2022).

Since real values or EID of nonlinearities and disturbances are unknown, it is not easy to directly analyze and design disturbance-estimation performance for the DOB, ESO, and EID estimator. Hence, a critical challenge for the aforementioned active disturbance-rejection control methods is how to achieve a performance-oriented optimization design.

Recently, an auxiliary variable was introduced to facilitate performance analysis of EID-based linear control systems (Yu et al., 2022). Further, a nonlinear function gain was adopted to improve dynamic performance (Yu et al., 2024). However, some key parameters were tuned manually, which was a bit tedious. Moreover, the performance degraded to the level of a linear method for the case where the noise and auxiliary variable shared the same order of magnitude. Therefore, how to achieve an

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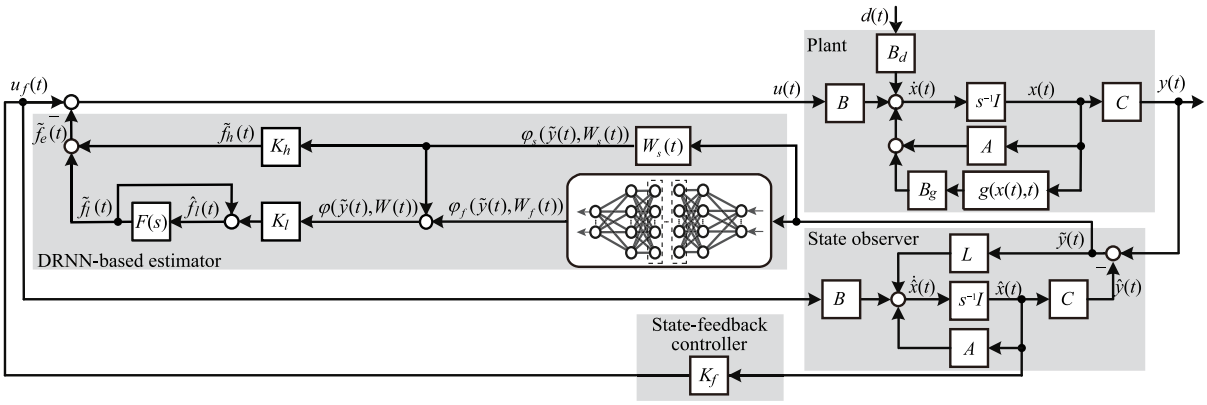


Fig. 1. Configuration of NN-assisted disturbance-rejection control system. It is constructed by inserting a DRNN-based estimator into a conventional control system (state observer and state feedback).

effective disturbance-rejection optimization design for nonlinear systems awaits a breakthrough.

In recent years, to empower dynamic systems with self-learning and adaptation abilities, neural networks have been widely used in control engineering. There are indirect and direct NN-assisted control designs. Indirect ones use NNs to estimate nonlinearities and these estimates are subsequently utilized to design control commands (Merabet et al., 2023). Direct ones use NNs to formulate control commands. Various techniques were developed to constrain NN weight estimates within bounded ranges of the ideal values of the weights. However, these techniques lacked convergence and closed-loop performance analysis (Xu & Wu, 2024). To improve learning efficacy, error back-propagation techniques were typically used (Abdollahi et al., 2006). However, the closed-loop stability cannot be rigorously analyzed mathematically (Emami et al., 2022). On the other hand, it could be costly to collect adequate data for NN training under unknown disturbances. Therefore, it is imperative to explore new ideas for NN-assisted disturbance-rejection control with performance guarantees.

Motivated by the aforementioned challenges, with only the system output, an NN-assisted disturbance-rejection control scheme is developed in this paper, which is based on a two-stage optimization-based learning control scheme. The core idea is to rapidly drive the disturbance-rejection error into a bounded range in the first rough learning stage and then to further online optimize the performance by error back-propagation algorithm in the second refined learning stage. The highlights lie in the following aspects:

- An NN-assisted EID estimator is tailored for performance-oriented control design, of which a dynamic residual NN (DRNN) including a residual term and a feedforward NN is incorporated.
- An adaptive law with exponential convergence is presented for the residual term of the rough learning stage, which takes (Abdollahi et al., 2006; Yu & Wu, 2024) as special cases.
- The closed-loop control performance with integrated disturbance-rejection performance is formulated and a hierarchical online learning algorithm is presented.
- Experimental comparisons of a rotary inverted pendulum system with four other methods show superior fast disturbance-rejection performance.

In this paper, $\mathbb{R}^{m\times n}$ denotes the set of real matrices with m rows and n columns; I, 0 stand for dimensioned identity, zero matrix, respectively; $P > (\geq) 0$ means P is a symmetric, positive definite (semi-definite) matrix; $\mathrm{tr}(A)$, $\mathrm{rank}(A)$, $\lambda_{\max}(A)$, and $\lambda_{\min}(A)$ denote the trace, rank, maximum eigenvalue, and minimum eigenvalue of the matrix A, respectively; $\mathrm{diag}\{\cdot\}$ denotes a diagonal matrix; $\|x(t)\|$ denotes the 2-norm of a vector; and time t is omitted for compactness when the context is clear.

2. System configuration

Consider a nonlinear system with a disturbance

$$\begin{cases}\dot{x}(t) = Ax(t) + Bu(t) + B_g g(x(t), t) + B_d d(t)\\ y(t) = Cx(t)\end{cases} \tag{1}$$

where $x(t) \in \mathbb{R}^n$ is the state; $u(t) \in \mathbb{R}^{n_u}$, $y(t) \in \mathbb{R}^{n_y}$ are the control input, system output, respectively; A, B, B_g, and C are dimensioned real constant matrices; $g(x(t), t) \in \mathbb{R}^{n_g}$ is the system nonlinearity; and $d(t) \in \mathbb{R}^{n_d}$ is the unknown disturbance with B_d being its input matrix.

Remark 1. If $\mathrm{rank}([B, B_d]) \neq \mathrm{rank}(B)$ holds, i.e., matrices B and B_d have different ranges, then the disturbance $d(t)$ is mismatched; otherwise, it is matched (Castanos & Fridman, 2006).

Assumption 1. (A, B, C) is controllable and observable.

Assumption 2. With known matrix R, for any $x(t)$, $z(t)$, the nonlinearity $g(x(t), t)$ satisfies $g(0, t) = 0$ and

$$\|g(x(t), t) - g(z(t), t)\| \leq \|R[x(t) - z(t)]\|. \tag{2}$$

Assumption 3. The disturbance $d(t)$ satisfies

$$\|d(t)\| \leq d_m, \quad t \geq 0 \tag{3}$$

where $d_m > 0$ is an unknown upper bound.

Assumption 1 is standard, which ensures the internal stability of the system (1) (Levine, 1996); Assumption 2 captures a variety of nonlinear systems (Guo & Chen, 2005); and Assumption 3 usually holds in practice.

The configuration of an NN-assisted disturbance-rejection control system is shown in Fig. 1, which is constructed by inserting a DRNN-based estimator into a conventional control system (state observer and state feedback). The main role of each component is as follows: DRNN-based estimator estimates an EID; the state observer provides the state estimates; and the state feedback controller stabilizes the overall system. The details will be explained in the following.

Under Assumption 1, the state observer and state feedback controller are designed as below. A full-order state observer is designed as

$$\dot{\hat{x}}(t) = A\hat{x}(t) + Bu_f(t) + L[y(t) - \hat{y}(t)],\ \hat{y}(t) = C\hat{x}(t) \tag{4}$$

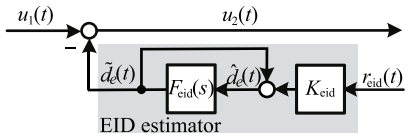


Fig. 2. EID estimator.

where $\hat{x}(t)$, $\hat{y}(t)$ denote the state, output, respectively; L is the observer gain to be determined; and $u_f(t)$ is an original feedback control law given by

$$u_f(t) = K_f\hat{x}(t) \tag{5}$$

with the feedback gain K_f to be determined.

Denote $\tilde{f}_e(t)$ as the output of the DRNN-based estimator and integrating $\tilde{f}_e(t)$ into $u_f(t)$ yields

$$u(t) = u_f(t) - \tilde{f}_e(t). \tag{6}$$

3. DRNN-based estimator

A brief review of the EID approach is recalled. An EID estimate $\hat{d}_e(t)$ is constructed by $\hat{d}_e(t) = K_{\text{eid}}r_{\text{eid}}(t) + u_1(t) - u_2(t)$, where K_{eid} is the control gain of an EID estimator; and $r_{\text{eid}}(t)$, $u_1(t)$, $u_2(t)$ denote the input, an original control law, a new control law with an EID compensation, respectively. Since $\hat{d}_e(t)$ is a function of itself, a filter $F_{\text{eid}}(s)$ is introduced to solve the causality problem (Yin, Shi, She, Wang, 2024) as shown in Fig. 2. Compared with a low-pass filter, a general filter that includes both a low-pass term and a direct through term provides more flexibility for control design and system stability by adding stable zeros (Du et al., 2020). However, the direct through term also may introduce significant noise.

The developed DRNN-based estimator in Fig. 1 can be viewed as a combination of a DRNN and an EID estimator with a general filter, which is tailored for achieving both stability and noise reduction. The DRNN including a simple residual term and a feedforward NN of a multilayer perceptron form is incorporated to fully exploit the information on the nonlinearity and disturbance. The form of the general filter is different from the one in Du et al. (2020). To be specific, the low-pass term and direct through term of the general filter are blended for Du et al. (2020), while for the developed structure, the direct through term (corresponding to the gain K_h) is separated from the low-pass term (i.e., the low-pass filter $F(s)$). Consequently, only the output of the residual term passes by the direct through term to improve stability, while the output of the feedforward NN does not pass by the direct through term to reduce noise. Moreover, the outputs of the residual term and feedforward NN are both filtered by the low-pass term to eliminate noise.

Denote the output of the DRNN as

$$\varphi(\tilde{y}(t), W(t)) = \varphi_s(\tilde{y}(t), W_s(t)) + \varphi_f(\tilde{y}(t), W_f(t)) \tag{7}$$

where $\tilde{y}(t)$ is the output error given by

$$\tilde{y}(t) = y(t) - \hat{y}(t); \tag{8}$$

$W_s(t)$, $W_f(t)$ are the weights of the residual term, feedforward NN, respectively; $W(t) = \{W_s(t), W_f(t)\}$; and $\varphi_s(\tilde{y}(t), W_s(t)) \in \mathbb{R}^{n_y}$, $\varphi_f(\tilde{y}(t), W_f(t)) \in \mathbb{R}^{n_y}$ are the outputs of the residual term, feedforward NN, also the inputs of the high-pass, low-pass sessions of the general EID estimator, respectively.

A state-space description of the low-pass filter $F(s)$ is

$$\dot{x}_F(t) = A_F x_F(t) + B_F\hat{f}_l(t), \quad \tilde{f}_l(t) = C_F x_F(t) \tag{9}$$

where $x_F(t)$ is the state; $\hat{f}_l(t)$, $\tilde{f}_l(t)$ are unfiltered, filtered EID estimates, respectively; and $(A_F,\ B_F,\ C_F)$ are system parameters. Then we have

$$\tilde{f}_e(t) = \tilde{f}_h(t) + \tilde{f}_l(t),\ \tilde{f}_h(t) = K_h\varphi_s(\tilde{y}(t), W_s(t)) \tag{10}$$

$$\tilde{f}_l(t) = \hat{f}_l(t) - K_l\varphi(\tilde{y}(t), W(t)) \tag{11}$$

where the gains $K_l \in \mathbb{R}^{n_u\times n_y}$ and $K_h \in \mathbb{R}^{n_u\times n_y}$ are to be calculated.

Combining (9) to (11) yields

$$\begin{cases}\dot{x}_F(t) = A_e x_F(t) + B_F K_l\varphi(\tilde{y}(t), W(t))\\ \tilde{f}_e(t) = C_F x_F(t) + K_h\varphi_s(\tilde{y}(t), W_s(t))\end{cases} \tag{12}$$

where

$$A_e = A_F + B_F C_F. \tag{13}$$

4. Dynamics of closed-loop system

To describe the closed-loop system dynamics, define

$$\tilde{x} = x - \hat{x},\ \ \psi = [\tilde{x}^T\ x_F^T]^T,\ \ x_c = [\psi^T\ \hat{x}^T]^T \tag{14}$$

$$A_L = A - LC,\ \ A_K = A + BK_f,\ \ K_e = [K_l^T\ \ K_h^T]^T. \tag{15}$$

From (1) and (4), we have

$$\dot{\tilde{x}} = A_L\tilde{x} - B\tilde{f}_e + B_g g(x,t) + B_d d(t),\ \ \tilde{y} = C\tilde{x} \tag{16}$$

where $\tilde{x}$ is given in (14).

According to (7), (12), and (16), we have

$$\begin{cases}\dot{\psi} = A_\psi\psi + B_s^\psi\varphi_s + \bar{B}_g g + B_f^\psi\varphi_f + \bar{B}_d d\\ \tilde{y} = C_\psi\psi\end{cases} \tag{17}$$

where ψ is given in (14), $A_\psi = [A_L\ \ -BC_F; 0\ \ A_e]$, $B_s^\psi = B_\psi K_e$, $B_\psi = [-B\ 0; 0\ B_F]$, $\bar{B}_g = [B_g^T\ 0]^T$, $B_f^\psi = [0\ (B_F K_l)^T]^T$, $\bar{B}_d = [B_d^T\ 0]^T$, $C_\psi = [C^T\ 0]^T$.

Substituting (5) into (4) gives

$$\dot{\hat{x}} = A_K\hat{x} + L\tilde{y}. \tag{18}$$

A combination of (17) and (18) gives

$$\dot{x}_c = A_c x_c + \tilde{B}_s\varphi_s + \tilde{B}_g g + \tilde{B}_f\varphi_f + \tilde{B}_d d \tag{19}$$

where x_c is given in (14), $A_c = [A_\psi\ \ 0; LC_\psi\ \ A_K]$, $\tilde{B}_s = [(B_s^\psi)^T\ \ 0]^T$, $\tilde{B}_g = [\bar{B}_g^T\ \ 0]^T$, $\tilde{B}_f = [(B_f^\psi)^T\ \ 0]^T$, $\tilde{B}_d = [\bar{B}_d^T\ \ 0]^T$.

5. Hierarchical online learning control

The plant output usually indicates the control performance of a closed-loop system. The output $y(t)$ of the nonlinear system (1) is rewritten as

$$y = \hat{y} + \tilde{y} = C\hat{x} + \tilde{y}. \tag{20}$$

We have the following analysis and design for disturbance-rejection and closed-loop control.

5.1. Analysis and design of disturbance rejection

Aimed at minimizing the influence of the disturbance and nonlinearity on the output estimate $\hat{y}(t)$, the vector 2-norm of $\tilde{y}(t)$ is used since $\tilde{y}(t)$ can be viewed as a disturbance to the system composed of (18) and (20). It leads to indirect disturbance rejection on the output $y(t)$ and gives D1 design problem:

$$\text{D1}: \min_{\{L,K_e,K_f,W(t)\}} J_{\tilde{y}}, \quad \text{where } J_{\tilde{y}} = \frac{1}{2}\|\tilde{y}(t)\|^2. \tag{21}$$

A two-stage online learning control scheme is developed to minimize $J_{\tilde{y}}$. Specifically, the first rough stage is a fast dynamic control with exponential convergence, and the second refined stage is an online optimization built on the first rough stage.

5.1.1. Rough learning control of residual term

The rough learning control aims to rapidly drive $\tilde{y}(t)$ into a neighborhood of the optimum point (i.e., the origin). A simple structure is adopted for the residual term by

$$\varphi_s(\tilde{y}(t), W_s(t)) = W_s(t)\tilde{y}(t). \tag{22}$$

Further, an adaptive learning law is designed.

Theorem 1. *With Assumptions 1 to 3, the closed-loop system of Fig. 1 is input-to-state stable and the performance index $J_{\tilde{y}}$ of (21) exponentially decays to a neighborhood of the origin if the conditions outlined below are satisfied.*

(1) The adaptive weight is taken as

$$W_s = -\|\tilde{y}(t)\|^{\mu} \int_0^t e^{-\Phi(t-\tau)} \|\tilde{y}(\tau)\|^{\mu} \Gamma \tilde{y}(\tau) \tilde{y}^T(\tau) d\tau \tag{23}$$

where $\Gamma > 0$, $\Phi \geq 0$, and μ is a bounded constant or function.

(2) The output $\varphi_f(\tilde{y}(t), W_f(t))$ of the feedforward NN is bounded.

(3) The system $(A_\psi,\ B_s^\psi,\ C_\psi)$ is positive real. Further, the positive realness of the system $(A_\psi,\ B_s^\psi,\ C_\psi)$ is equivalent to the existence of matrices $Q_1 > 0$, N, and K_e such that the following equations hold:

$$A_\psi Q_1 + Q_1 A_\psi^T = -N^T N, \quad B_\psi K_e = Q_1 C_\psi^T. \tag{24}$$

(4) For Q_1 satisfying (24), a given positive scalar γ, there exist matrices $Q_2 > 0$, U, and positive scalars ε_2, ε_4 such that the following inequality holds:

$$\begin{bmatrix} S_1 & \gamma Q_1 C_\psi^T L^T & \varepsilon_2 \bar{B}_g & Q_1 \bar{R}^T \\ \gamma L C_\psi Q_1 & \gamma^2 S_2 & 0 & \gamma Q_2 R^T \\ \varepsilon_2 \bar{B}_g^T & 0 & -\varepsilon_2 & 0 \\ \bar{R} Q_1 & \gamma R Q_2 & 0 & -\varepsilon_2 \end{bmatrix} < 0 \tag{25}$$

where L is the observer gain, R is given by (2), $\bar{R} = [R\ \ 0]$, $S_1 = A_\psi Q_1 + Q_1 A_\psi^T + \varepsilon_4 I$, and $S_2 = AQ_2 + BU + (AQ_2 + BU)^T$. Moreover,

$$K_f = U Q_2^{-1}. \tag{26}$$

Proof. Denote $\sigma = -\int_0^t e^{-\Phi(t-\tau)} \|\tilde{y}\|^{\mu} \Gamma \tilde{y} \tilde{y}^T d\tau$, then

$$W_s = \|\tilde{y}\|^{\mu} \sigma^T, \ \ \dot{\sigma} = -\|\tilde{y}\|^{\mu} \Gamma \tilde{y} \tilde{y}^T - \Phi \sigma. \tag{27}$$

For the system of (19) and σ of (27), choose a Lyapunov functional

$$V = x_c^T P x_c + \mathrm{tr}(\sigma^T \Gamma^{-1} \sigma) \tag{28}$$

where $P = \mathrm{diag}\{P_1,\ P_2\} > 0$ is to be determined later. The derivative of V is

$$\begin{aligned} \dot{V} =& \psi^T \left(P_1 A_\psi + A_\psi^T P_1\right) \psi + \hat{x}^T \left(P_2 A_K + A_K^T P_2\right) \hat{x} \\ &+ 2\hat{x}^T P_2 L C_\psi \psi + 2\psi^T P_1 B_s^\psi \varphi_s + 2\psi^T P_1 B_f^\psi \varphi_f \\ &+ 2x_c^T P \tilde{B}_g g + 2\psi^T P_1 \bar{B}_d d + 2\mathrm{tr}(\sigma^T \Gamma^{-1} \dot{\sigma}). \end{aligned} \tag{29}$$

According to the positive real lemma (Khalil, 2002), for the positive real system $(A_\psi,\ B_s^\psi,\ C_\psi)$, there exists matrices $P_1 > 0$, M such that

$$P_1 A_\psi + A_\psi^T P_1 = -M^T M, \quad P_1 B_s^\psi = C_\psi^T. \tag{30}$$

Let $Q_1 = P_1^{-1}$, $Q_2 = P_2^{-1}$. Pre- and post-multiplying the first equality of (30) by Q_1 give (24), where $N = MQ_1$.

By Young's inequality, the indefinite terms in $\dot{V}$ of (29) can be bounded as follows:

$$2\psi^T P_1 B_f^\psi \varphi_f \leq \varepsilon_1^{-1} \psi^T P_1 P_1 \psi + \varepsilon_1 (B_f^\psi \varphi_f)^T B_f^\psi \varphi_f$$

$$2x_c^T P \tilde{B}_g g \leq \varepsilon_2 x_c^T P \tilde{B}_g (x_c^T P \tilde{B}_g)^T + \varepsilon_2^{-1} g^T g$$

$$2\psi^T P_1 \bar{B}_d d \leq \varepsilon_3^{-1} \psi^T P_1 P_1 \psi + \varepsilon_3 \left(\bar{B}_d d\right)^T \bar{B}_d d$$

where ε_1 to ε_3 are any positive scalars. By Assumption 2, we have $0 \leq -\varepsilon_2^{-1} g^T g + \varepsilon_2^{-1} x_c^T \tilde{R}^T \tilde{R} x_c$, where $\tilde{R} = \left[\bar{R}\ \ R\right]$, and $\bar{R}$, R are defined in the condition (4). Further, with W_s of (27), we have

$$\begin{aligned} \dot{V} \leq& x_c^T Z x_c - 2\mathrm{tr}(\sigma^T \Gamma^{-1} \Phi \sigma) \\ &+ \varepsilon_1 (B_f^\psi \varphi_f)^T B_f^\psi \varphi_f + \varepsilon_3 \left(\bar{B}_d d\right)^T \bar{B}_d d \end{aligned} \tag{31}$$

where $Z = X + \varepsilon_2 P \tilde{B}_g (P \tilde{B}_g)^T + \varepsilon_2^{-1} \tilde{R}^T \tilde{R}$, $X = [X_1\ \ \left(P_2 L C_\psi\right)^T; P_2 L C_\psi\ \ P_2 A_K + A_K^T P_2]$, $X_1 = P_1 A_\psi + A_\psi^T P_1 + \left(\varepsilon_1^{-1} + \varepsilon_3^{-1}\right) P_1 P_1$.

Note that d is bounded by Assumption 3 and that φ_f is bounded by the condition (3) of Theorem 1. If

$$Z < 0 \tag{32}$$

then the closed-loop system of Fig. 1 is input-to-state stable. Further, by combining (28) and (31), we have

$$\dot{V} \leq -\beta_1 V + \beta_2$$

where $\beta_1 = \min\{2, \lambda_{\min}(-Z)/\lambda_{\max}(P)\}$ and $\beta_2 = \varepsilon_1 \lambda_{\max}((B_f^\psi)^T B_f^\psi) \bar{\varphi}_f^2 + \varepsilon_3 \lambda_{\max}(\bar{B}_d^T \bar{B}_d) d_m^2$ with $\bar{\varphi}_f$ being the upper bound of φ_f. By the comparison lemma of Khalil (2002), we have

$$V \leq e^{-\beta_1 t} V(0) + \frac{\beta_2}{\beta_1} \left(1 - e^{-\beta_1 t}\right).$$

To reveal the relation between $J_{\tilde{y}}$ and V, note that $2J_{\tilde{y}} = (C\hat{x})^T (C\hat{x}) = (C_\psi \psi)^T (C_\psi \psi) \leq \lambda_{\max}(C_\psi^T C_\psi) \psi^T \psi$. An expansion of the first term on the right-hand side of (28) by the partitioned matrix $P = \mathrm{diag}\{P_1,\ P_2\}$ yields $\lambda_{\min}(P_1) \psi^T \psi \leq \psi^T P_1 \psi \leq V$.

Hence, we have

$$J_{\tilde{y}} \leq \frac{\lambda_{\max}(C^T C)}{2\lambda_{\min}(P_1)} \left[e^{-\beta_1 t} V(0) + \frac{\beta_2}{\beta_1} \left(1 - e^{-\beta_1 t}\right)\right].$$

The performance index $J_{\tilde{y}}$ of (21) exponentially converges to a neighborhood of the origin.

By an application of Schur complement argument to $Z < 0$ of (32), and then pre- and post-multiplying the resultant inequality by $\mathrm{diag}\{Q_1, \gamma Q_2, I, I\}$, we have (25) by defining $\varepsilon_4 = \varepsilon_1^{-1} + \varepsilon_3^{-1}$. This completes the proof. □

Recurring to the well-known Kalman-Yakubovich-Popov lemma (Khalil, 2002), an equivalent statement to the condition (3) of Theorem 1 is obtained.

Proposition 1. *The system $(A_\psi,\ B_s^\psi,\ C_\psi)$ is positive real if there exist matrices $Q_1 > 0$, $Q_3 \geq 0$, and K_e such that the following inequality holds:*

$$\begin{bmatrix} Q_1 A_\psi^T + A_\psi Q_1 + Q_3 & B_\psi K_e - Q_1 C_\psi^T \\ K_e^T B_\psi^T - C_\psi Q_1 & 0 \end{bmatrix} \leq 0. \tag{33}$$

Remark 2. Since a complex feedforward NN will be designed for the second stage, the residual term of the first stage can be simple. Moreover, a simple weight simplifies system analysis. For the special case $\mu = 0$, the weight updated law $\dot{W}_s = -\tilde{y}\tilde{y}^T \Gamma$ with $\Phi = 0$ resembles (Abdollahi et al., 2006), which was calculated by the error backpropagation algorithm using static system gradient.

5.1.2. Refined learning control of feedforward NN

Building on the rough learning control of the first stage, an online optimization algorithm is used to further refine the performance index $J_{\tilde{y}}$ of (21).

Without loss of generality, consider the following feedforward NN composed of an input layer, one hidden layer with n_n neurons, and an output layer.

$$\varphi_f(\tilde{y}(t), W_f(t)) = W_f^2(t)h(W_f^1(t)\tilde{y}(t)) \tag{34}$$

where $W_f^1(t) \in \mathbb{R}^{n_n \times n_y}$ is the connecting weights between the input layer and hidden layer, and $W_f^2(t) \in \mathbb{R}^{n_y \times n_n}$ is the connecting weights between the hidden layer and the output layer. Without loss of generality, all the neurons are taken in the form of hyperbolic tangent activation functions, i.e., tanh(·).

Theorem 2. *On the basis of completing the rough learning control developed in Theorem 1 (which ensures $J_{\tilde{y}}$ of (21) exponentially decays to a neighborhood of the origin), the global (or a local) optimum of the disturbance-rejection performance index $J_{\tilde{y}}$ of (21) can be obtained if the optimal weight is inside (or outside) the range $\|W_f^2(t)\| < W_m^2$; moreover, the output $\varphi_f(\tilde{y}(t), W_f(t))$ (34) of the feedforward NN is bounded when the following learning laws are adopted:*

$$\dot{W}_f^1 = -\eta_1 \kappa_1(\tilde{y}) \tag{35}$$

$$\dot{W}_f^2 = \begin{cases} -\eta_2 \kappa_2(\tilde{y}), & \text{for } \|W_f^2(t)\| < W_m^2 \\ 0, & \text{for } \|W_f^2(t)\| \geq W_m^2 \end{cases} \tag{36}$$

where η_1, η_2 are learning rates; W_m^2 is a given bound; $\kappa_1(\tilde{y}) = (\tilde{y}^T \frac{\partial \tilde{y}}{\partial \varphi_f} W_f^2 \Xi)^T \tilde{y}^T$; $\kappa_2(\tilde{y}) = (\tilde{y}^T \frac{\partial \tilde{y}}{\partial \varphi_f})^T h^T(W_f^1 \tilde{y})$; $\Xi = I - h(W_f^1 \tilde{y}) h^T(W_f^1 \tilde{y})$; and $\frac{\partial \tilde{y}}{\partial \varphi_f} = C_\psi (e^{A_\psi t} - I) A_\psi^{-1} B_f^\psi$.

Proof. Choose the disturbance-rejection performance index $J_{\tilde{y}}$ of (21) as the cost function of the feedforward NN. Then the derivative of $J_{\tilde{y}}$ is

$$\dot{J}_{\tilde{y}} = \left(\frac{\partial J_{\tilde{y}}}{\partial W_f^1}\right)^T \dot{W}_f^1 + \left(\frac{\partial J_{\tilde{y}}}{\partial W_f^2}\right)^T \dot{W}_f^2 \tag{37}$$

By the chain rule, we have

$$\frac{\partial J_{\tilde{y}}}{\partial W_f^1} = \frac{\partial J_{\tilde{y}}}{\partial \tilde{y}} \frac{\partial \tilde{y}}{\partial \varphi_f} \frac{\partial \varphi_f}{\partial (W_f^1 \tilde{y})} \tilde{y}^T = (\tilde{y}^T \frac{\partial \tilde{y}}{\partial \varphi_f} W_f^2 \Xi)^T \tilde{y}^T \tag{38}$$

$$\frac{\partial J_{\tilde{y}}}{\partial W_f^2} = \frac{\partial J_{\tilde{y}}}{\partial \tilde{y}} \frac{\partial \tilde{y}}{\partial \varphi_f} h^T(W_f^1 \tilde{y}) = (\tilde{y}^T \frac{\partial \tilde{y}}{\partial \varphi_f})^T h^T(W_f^1 \tilde{y}) \tag{39}$$

where Ξ is defined in Theorem 2. To calculate $\frac{\partial \tilde{y}}{\partial \varphi_f}$, note that $\tilde{y} = C_\psi e^{A_\psi t}\psi(0) - \int_0^t C_\psi e^{A_\psi}(t-\tau) B_s^\psi \varphi_s(\tau) d\tau - \int_0^t C_\psi e^{A_\psi}(t-\tau)\bar{B}_g g(\tau) d\tau - \int_0^t C_\psi e^{A_\psi}(t-\tau)\bar{B}_d d(\tau) d\tau - \int_0^t C_\psi e^{A_\psi}(t-\tau) B_f^\psi \varphi_f(\tau) d\tau$ from (17). Treating other variables except for φ_f as constants, we have $\frac{\partial \tilde{y}}{\partial \varphi_f}$ given in Theorem 2.

Note $\kappa_1(\tilde{y})$ of (35) and $\kappa_2(\tilde{y})$ of (36) equal to $\frac{\partial J_{\tilde{y}}}{\partial W_f^1}$ of (38) and $\frac{\partial J_{\tilde{y}}}{\partial W_f^2}$ of (39), respectively. Therefore, we have

$$\dot{W}_f^1 = -\eta_1 \frac{\partial J_{\tilde{y}}}{\partial W_f^1} \tag{40}$$

$$\dot{W}_f^2 = \begin{cases} -\eta_2 \frac{\partial J_{\tilde{y}}}{\partial W_f^2}, & \text{for } \|W_f^2(t)\| < W_m^2 \\ 0, & \text{for } \|W_f^2(t)\| \geq W_m^2. \end{cases} \tag{41}$$

Then for the case $\|W_f^2(t)\| < W_m^2$, we have

$$\dot{J}_{\tilde{y}} = -\eta_1 \left(\frac{\partial J_{\tilde{y}}}{\partial W_f^1}\right)^T \frac{\partial J_{\tilde{y}}}{\partial W_f^1} - \eta_2 \left(\frac{\partial J_{\tilde{y}}}{\partial W_f^2}\right)^T \frac{\partial J_{\tilde{y}}}{\partial W_f^2} \leq 0 \tag{42}$$

where $\dot{J}_{\tilde{y}} = 0$ only if both $\frac{\partial J_{\tilde{y}}}{\partial W_f^1} = 0$ and $\frac{\partial J_{\tilde{y}}}{\partial W_f^2} = 0$; and for the case $\|W_f^2(t)\| \geq W_m^2$, we have

$$\dot{J}_{\tilde{y}} = -\eta_1 \left(\frac{\partial J_{\tilde{y}}}{\partial W_f^1}\right)^T \frac{\partial J_{\tilde{y}}}{\partial W_f^1} \leq 0. \tag{43}$$

Further, according to (38), (39), and $\frac{\partial \tilde{y}}{\partial \varphi_f}$ given in Theorem 2, $\tilde{y} = 0$ when both $\frac{\partial J_{\tilde{y}}}{\partial W_f^1} = 0$ and $\frac{\partial J_{\tilde{y}}}{\partial W_f^2} = 0$. So, if an optimal weight belongs to the range $\|W_f^2(t)\| < W_m^2$, the global minimum of the disturbance-rejection performance index $J_{\tilde{y}}$ of (21) can be obtained. Otherwise, only a local minimum is obtained.

If $\|W_f^2(t)\| \geq W_m^2$, $W_f^2(t)$ would be constrained by the boundary because $\dot{W}_f^2 = 0$. Consequently, $W_f^2(t)$ is bounded. Since bounded active functions tanh(·) are taken for the feedforward NN, $\varphi_f(\tilde{y}(t), W_f(t))$ of (34) is also bounded. This completes the proof. □

Remark 3. To improve the generalization and lower the complexity of the weights, a regularization term can be introduced into the cost function as follows:

$$J_r = \frac{1}{2}\|\tilde{y}\|^2 + \frac{\rho_1}{2\eta_1} \mathrm{tr}\left((W_f^1)^T W_f^1\right) + \frac{\rho_2}{2\eta_2} \mathrm{tr}\left((W_f^2)^T W_f^2\right)$$

where η_1, η_2 are the learning rates as given in Theorem 2; and ρ_1, ρ_2 are small positive numbers. Following the similar procedures, with W_m^2, $\kappa_1(\tilde{y})$, and $\kappa_2(\tilde{y})$ defined in Theorem 2, we have the same result as (Yu & Wu, 2024):

$$\dot{W}_f^1 = -\eta_1 \kappa_1(\tilde{y}) - \rho_1 W_f^1 \tag{44}$$

$$\dot{W}_f^2 = \begin{cases} -\eta_2 \kappa_2(\tilde{y}) - \rho_2 W_f^2, & \text{for } \|W_f^2(t)\| < W_m^2 \\ 0, & \text{for } \|W_f^2(t)\| \geq W_m^2. \end{cases} \tag{45}$$

5.2. Analysis and design of closed-loop system

The signal 2-norm of $y(t)$ is used to denote the closed-loop control performance, which gives D2 design problem:

$$\mathtt{D2}: \min_{\{L, K_e, K_f, W(t)\}} \frac{1}{2}\|y\|^2, \quad \text{where } J_y = \frac{1}{2}\|y\|^2. \tag{46}$$

The relation of y and $\tilde{y}$ is clear by (20). For the two-objective design problem (D1 and D2), when $J_{\tilde{y}}$ is minimized, J_y can be minimized by minimizing the H_∞ norm of the system (A_K, L, C, I) composed of (18) and (20). So, by the bounded real lemma, we have the following theorem.

Theorem 3. *D2 design problem of (46) can be equivalently solved by*

$$D3: \min_{\{L, K_f\}} \chi$$
$$s.t. \begin{bmatrix} S_2 & L & Q_2 C^T \\ L^T & -\chi I & I \\ C Q_2 & I & -\chi I \end{bmatrix} < 0 \tag{47}$$

where χ is a positive scalar; and S_2, Q_2 are defined in the condition (4) of Theorem 1.

6. Hierarchical online learning algorithm

With the above preparation, the overall control design of the closed-loop system in Fig. 1 is to simultaneously solve D1 of (21) and D3 of (47). To this end, Theorems 1 and 2 are presented for D1 and Theorem 3 for D3. Specifically, (23) of Theorem 1 is to determine $W_s(t)$, and (24) (or (33)), (25) are bilinear constraints

on (L, K_e, K_f); (35) and (36) of Theorem 2 are to calculate $W_f(t)$; and the inequality of (47) of Theorem 3 is a linear constraint on (L, K_f).

To lower the computational burden, predesigning L renders both (24) (or (33)), (25) are linear in K_e and K_f. In other words, with L predesigning, (24) (or (33)), (25) simplify to linear constraints, and thus existing tools like YALMIP (Tutuncu et al., 2003) can be used. On the other hand, predesigning L does not inherently compromise the control performance because the design degrees of freedom of $W_f(t)$ and K_f can be used to achieve a minimization objective. Furthermore, methods such as the linear quadratic regulation (LQR) and pole placement can be used to predesign L such that A_L of (16) is stable.

Algorithm 1 Design of $K_e, K_f, W_s(t)$

Input: (A, B, C), R, (A_F, B_F, C_F), L;
Output: K_e, K_f, $W_s(t)$, $\tilde{y}(t)$;

1: **Start:**
2: Find feasible solutions to (24) (or (33)), (25), the inequality of (47) under the constraint that χ is minimized, obtain K_e, and calculate K_f from (26);
3: Obtain the weight $W_s(t)$ from (23) by selecting properly μ, Γ, and Φ;
4: **return** K_e, K_f, $W_s(t)$, $\tilde{y}(t)$;
5: **End.**

Algorithm 2 Design of $W_f(t)$.

Input: $\tilde{y}(t)$, $(A_\psi,\ B_s^\psi,\ C_\psi)$;
Output: $W_f(t)$;

1: Initialize: Choose $W_f^1(0)$, $W_f^2(0)$ randomly;
2: **Start:**
3: Obtain $\dot{W}_f(t)$ from (35), (36) (or (44) and (45)) by selecting η_1, η_2, (ρ_1, ρ_2), and W_m^2;
4: Obtain the weight $W_f(t)$ from $\dot{W}_f(t)$;
5: **return** $W_f(t)$;
6: **End.**

Summarizing the core findings leads to Algorithms 1 and 2. By combining these algorithms, we have the hierarchical online learning algorithm, i.e., Algorithm 3.

Algorithm 3 Hierarchical Online Learning Algorithm

Input: Nominal parameters (A, B, C) of (1), matrix R of (2);
Output: K_e, K_f, $W_s(t)$, $W_f(t)$;

1: **Start:**
2: Design (A_F, B_F, C_F) such that A_e of (13) is stable;
3: Design L to make A_L of (15) stable;
4: Implement the roughly learning control by using Algorithm 1 and obtain $K_e, K_f, W_s(t)$;
5: **if** the control performance is satisfactory **then**
6: disconnect the refined learning control stage by letting $W_f(t) = 0$;
7: **else**
8: implement the refined learning control stage using Algorithm 2 and obtain $W_f(t)$;
9: **end if**
10: **return** K_e, K_f, $W_s(t)$, $W_f(t)$;
11: **End.**

Remark 4. The stability of the developed hierarchical online learning control system is ensured by Theorems 1 and 2. There are four conditions in Theorem 1. The first one is the developed adaptive law. The second one is a boundedness requirement for the output $\varphi_f(\tilde{y}(t), W_f(t))$ of the feedforward NN. The third one is a positive realness requirement for the system $(A_\psi,\ B_s^\psi,\ C_\psi)$ and the system can be viewed as a cascade of $(A_L,\ -B,\ C)$ and $(A_e,\ B_F K_l,\ C_F,\ K_h)$. Consequently, the third one can be satisfied by properly selecting $(A_F,\ B_F,\ C_F)$ and designing (L, K_e). The last one is a system stabilization condition about K_f. Furthermore, Theorem 2 guarantees the boundedness of feedforward NN output $\varphi_f(\tilde{y}(t), W_f(t))$ by the developed weight learning laws (35) and (36).

Remark 5. This paper is a significant improvement over (Yu & Wu, 2024) in four aspects: the nonlinearity of (2) is more general than the bounded nonlinearity of Yu and Wu (2024); the developed DRNN-based estimator has a general structure with more design flexibility than that of Yu and Wu (2024); with the general structure, an exponential convergence of disturbance-rejection performance is guaranteed for the rough learning stage of the developed method, while only a bounded stability was obtained in Yu and Wu (2024); the disturbance-rejection performance is analyzed for the refined learning stage, while it is missing in Yu and Wu (2024); lastly, the analysis and design for the closed-loop control performance are conducted, which is also missing in Yu and Wu (2024).

Remark 6. The developed method has a hybrid nature of theory-oriented and data-driven control. Specifically, some parameters are designed offline, while weights W_s and W_f are adapted online. Compared to purely data-driven control, the rough learning stage guarantees fast initial convergence by a simple residual term. Compared to purely theory-oriented methods, the disturbance-rejection performance is further refined online by a feedforward NN.

Remark 7. Some guides are presented to show how to select key offline parameters. In Algorithm 1, when $\Phi > 0$, for an equivalent version $\Phi^{-1}\dot{\sigma} = -\|\tilde{y}\|^\mu \Phi^{-1}\Gamma\tilde{y}\tilde{y}^T - \sigma$ of (23), treating $-\|\tilde{y}\|^\mu \Phi^{-1}\Gamma\tilde{y}\tilde{y}^T$ as the input and σ as the corresponding output, Φ can be viewed as the bandwidth of a low-pass filter. Accordingly, a proper Φ can be used to filter out the high-frequency noise. Moreover, $\Phi^{-1}\Gamma$ can be regarded as a control gain, and a large gain usually yields a fast dynamic performance. However, a large $\Phi^{-1}\Gamma$ also amplifies the noise component of $\tilde{y}$. In practice, Γ and Φ should be co-tuned to achieve a tradeoff between dynamic performance and noise suppression. As for the parameter γ of (25), it is introduced to tune the feedback gain K_f by adjusting the matrix Q_2. Techniques such as grid search and meta-heuristic search can be used to select γ. In Algorithm 2, zeros or too large initial values should not be selected for $W_f^1(0)$, $W_f^2(0)$ to avoid gradient explosion or vanishing gradients; η_1, η_2 determine the learning speed of the feedforward NN; and W_m^2 gives the allowable range of the weight $W_f^2(t)$, of which the selection needs some trial and error since excessively large W_m^2 may lead to overfitting due to measurement noise.

7. Experiment verification

A rotary inverted pendulum experiment platform of "Quanser QUBE-servo 3" is used to validate the developed method in practice. It is mainly composed of a direct-drive brushed DC motor, a rotary pendulum consisting of a controller, a rotary arm and a pendulum arm, two single-ended rotary encoders, and other components. The controller is a desktop computer, of which the core configuration includes Intel Core i7-13700H, 16G RAM, MATLAB R2022b, and Windows 11 operating system. The experiment setup is shown in Fig. 3.

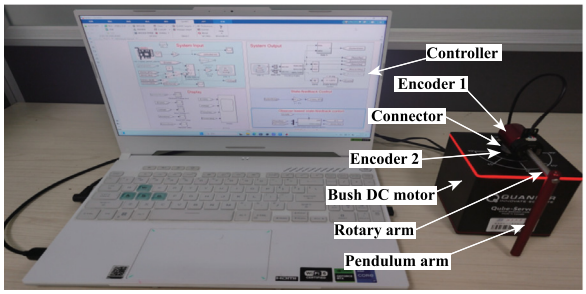


Fig. 3. Experiment setup of the rotary inverted pendulum.

Denote $\theta, \dot{\theta}, \ddot{\theta}$ and $\alpha, \dot{\alpha}, \ddot{\alpha}$ as the angle, velocities, acceleration of the rotary arm and pendulum arm, respectively; and v_m as the input voltage of the DC motor. The control objective is to balance the pendulum arm in the upright position by the rotary arm driven by the motor in the horizontal plane. To this end, two single-ended rotary encoders are used to measure θ and α.

The parameters with the subscript r (or p or m) indicate they are related to the rotary arm (or pendulum arm or motor). The related parameters are given. The arm lengths L_r and L_p are 0.085 m and 0.129 m, respectively; the arm masses m_r and m_p are 0.095 kg and 0.024 kg, respectively; the arm rotary inertia J_r and J_p are 2.288×10^{-4} kg m^2 and 1.331×10^{-4} kg m^2, respectively; the friction coefficients b_r and b_p are 0.006×10^{-1} N m s/rad and 0.005×10^{-2} N m s/rad, respectively; the resistance R_m of motor armature coil is 7.500 Ω; and the motor torque constant k_m is 0.042 N m/A.

A mathematical model of the rotary inverted pendulum is obtained (Misawa et al., 1996). $\ddot{\theta} = \frac{1}{k_2(\alpha)}\tau - \frac{k_3(\alpha)b_p}{J_p}\dot{\alpha} - \frac{(J_p\dot{\alpha}\sin 2\alpha + b_r)\dot{\theta}}{k_2(\alpha)} + g_{k1}$, and $\ddot{\alpha} = \frac{k_3(\alpha)}{J_p}\tau - \frac{[b_p + k_3(\alpha)k_1(\alpha)b_p]\dot{\alpha}}{J_p} - \frac{k_3(\alpha)(J_p\dot{\alpha}\sin 2\alpha + b_r)\dot{\theta}}{J_p} + g_{k2}$, where $k_1(\alpha) = \frac{m_p r l \cos\alpha}{J_p}$, $k_2(\alpha) = J_r + J_p \sin^2\alpha - \frac{(m_p r l \cos\alpha)^2}{J_p}$, $k_3(\alpha) = \frac{J_p k_1(\alpha)}{k_2(\alpha)}$, and $g_{k1} = -\frac{J_p}{k_2(\alpha)}\dot{\alpha}\dot{\theta}\sin 2\alpha - \frac{m_p r l}{k_2}\dot{\alpha}^2\sin\alpha + k_3\dot{\theta}^2\sin\alpha\cos\alpha$, $g_{k2} = -\frac{(m_p r l)^2}{J_r}\dot{\alpha}^2\sin\alpha + \dot{\theta}^2\sin\alpha\cos\alpha + k_3(\alpha)k_1(\alpha)\dot{\theta}^2\sin\alpha\cos\alpha$.

Let $x = [\theta, \alpha, \dot{\theta}, \dot{\alpha}]^T$ and $u = v_m$. Linearization of $k_1(\alpha)$, $k_2(\alpha)$, and $k_3(\alpha)$ at the equilibrium point ($\alpha = 0$) yields $k_1 = \frac{m_p r l}{J_p}$, $k_2 = \frac{J_r J_p - (m_p r l)^2}{J_p}$, and $k_3 = \frac{J_p m_p r}{J_r J_p - (m_p r l)^2}$.

Then we have

$$\begin{cases} A = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & a_1 & a_2 & a_3 \\ 0 & a_4 & a_5 & a_6 \end{bmatrix}, B = \begin{bmatrix} 0 \\ 0 \\ b_1 \\ b_2 \end{bmatrix}, B_g = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 1 & 0 \\ 0 & 1 \end{bmatrix} \\ C = \begin{bmatrix} 1 & 0 & 0.01 & 0 \\ 0 & 1 & 0 & 0.02 \end{bmatrix}, g = \begin{bmatrix} g_1 \\ g_2 \end{bmatrix} \end{cases} \tag{48}$$

where $a_1 = 152.01$, $a_2 = -6.09$, $a_3 = -0.50$, $a_4 = 264.31$, $a_5 = -6.02$, $a_6 = -0.51$, $b_1 = 56.71$, $b_2 = 56.05$, $g_1 = -1.35\dot{\alpha}\dot{\theta}\sin 2\alpha - 1.33\dot{\alpha}^2\sin\alpha + 0.67\dot{\theta}^2\sin 2\alpha$, and $g_2 = -1.32\dot{\alpha}^2\sin\alpha + 1.16\dot{\theta}^2\sin 2\alpha$. Consider the case where $\alpha \in [-10°, 10°]$, $\dot{\theta} \in [-18°/s, 18°/s]$, and $\dot{\alpha} \in [-6°/s, 6°/s]$, then the matrix R defined in (2) is $[R_1^T\ R_2^T]^T$, where $R_1 = [0\ \ 0.18\ \ 0.18\ \ 0.18]$ and $R_2 = [0\ \ 0.21\ \ 0.24\ \ 0.05]$.

To validate the developed method, a common scenario with the nonzero initial state $x(0) = [0\ \ 8°\ \ 0\ \ 0]^T$ is set for reproducibility and the following voltage disturbance with $B_d = B$ is added.

$$d(t) = \begin{cases} 3\sin 2\pi t, & 10\text{ s} \le t \le 20\text{ s} \\ 5, & 25\text{ s} < t \le 35\text{ s} \\ 0, & \text{otherwise.} \end{cases} \tag{49}$$

The parameters of the filter are selected as

$$A_F = -1.05,\ B_F = 1,\ C_F = 1. \tag{50}$$

The LQR is used to design L such that A_L is Hurwitz. The corresponding weighting matrices are $Q_L = \text{diag}\{1, 1, 100, 100\} \times 10^5$, $R_L = 1$, which give

$$L = \begin{bmatrix} 303.98 & 11.934 & 1967.5 & -672.04 \\ 4.7230 & 284.25 & -622.98 & 2533.5 \end{bmatrix}^T. \tag{51}$$

Further, letting $\gamma = 10^3$ and solving the D2 minimization problem of (46) under the constraints (24), (25), and (47) give $\chi = 2.7291 \times 10^2$ and

$$\begin{cases} K_l = [-0.8033 \quad -0.7635] \times 10^{-5} \\ K_h = [-0.1099 \quad -0.0803] \times 10^{-4} \\ K_f = [13.213 \quad -56.111 \quad 3.3594 \quad -4.5744]. \end{cases} \tag{52}$$

The cost time of solving the three inequalities of Algorithm 1 is about 0.3 s. The parameters of the residual term and feedforward NN are $\mu = 0$, $\Gamma = \text{diag}\{4, 0.01\} \times 10^{12}$, $\Phi = \text{diag}\{10, 100\}$, $W_f^1(0), W_f^2(0) \in [-0.01, 0.01]$, $\eta_1 = \eta_2 = 10^{11}$, $\rho_1 = \rho_2 = 100$, $W_m^2 = 600$.

7.1. Experiment results and analysis

To demonstrate the control performance of the developed method, the rough learning control works from start and the refined learning control is applied from the 5th second.

The system response in Fig. 4 shows the rotary inverted pendulum control system is robustly stabilized. With only the rough learning control, the system converges within 0.5 s as shown by Fig. 4(e), demonstrating rapid convergence under the combined effects of the nonlinearity, nonzero initial state, and noise; together with the second learning control, the unknown disturbance is fast and well compensated by the developed DRNN-based estimator as shown by Fig. 4(d) and (f); and the weight $W_s(t)$ adaptively varies in response to the nonzero initial state, noise, and disturbance as shown by Fig. 4(b).

7.2. Experiment comparison with other methods

Four other methods are used to compare with the developed method, which are the control methods without the EID estimator (W/O EID), with the EID estimator (She et al., 2008), with the improved EID (IEID) estimator (Du et al., 2020), and with the modified EID (MEID) estimator (Yin, Shi, She, Xie et al., 2024). For a fair comparison, the parameters (A_F, B_F, C_F), L, and K_f for the above methods are the same as the develop method, as specified by (50), (51), and (52). The gain K_H is chosen as 1.5 for the IEID estimator (Du et al., 2020) since a larger K_H would cause a worse noise impact. The comparative results are shown in Fig. 5.

According to the control input of Fig. 5(a), noise suppression is similar for the proceeding five methods. For the same nonzero initial state, even though the peak value of the control input is smaller than the IEID estimator, the developed DRNN-based estimator has a better dynamic performance for the rough learning control. Moreover, for the same disturbance (49), the developed DRNN-based estimator has the best control performance as shown by Fig. 5(b) and (c).

8. Conclusion

A hierarchical online learning control scheme with performance consideration was presented for nonlinear systems with unknown disturbances in this paper, of which an adaptive DRNN was integrated into an EID estimator. First, a disturbance-

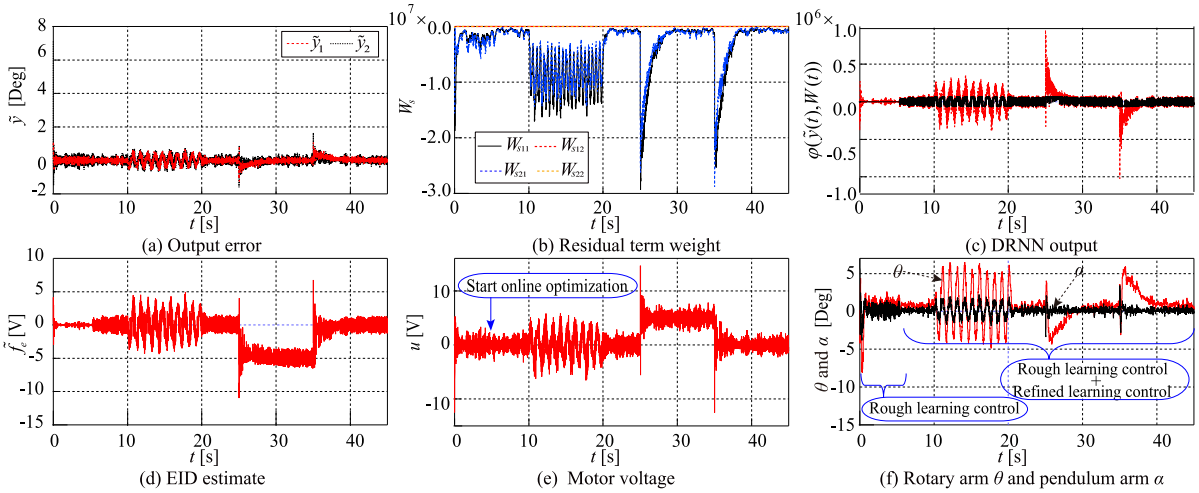


Fig. 4. Experimental results of DRNN-based rotary inverted pendulum control system.

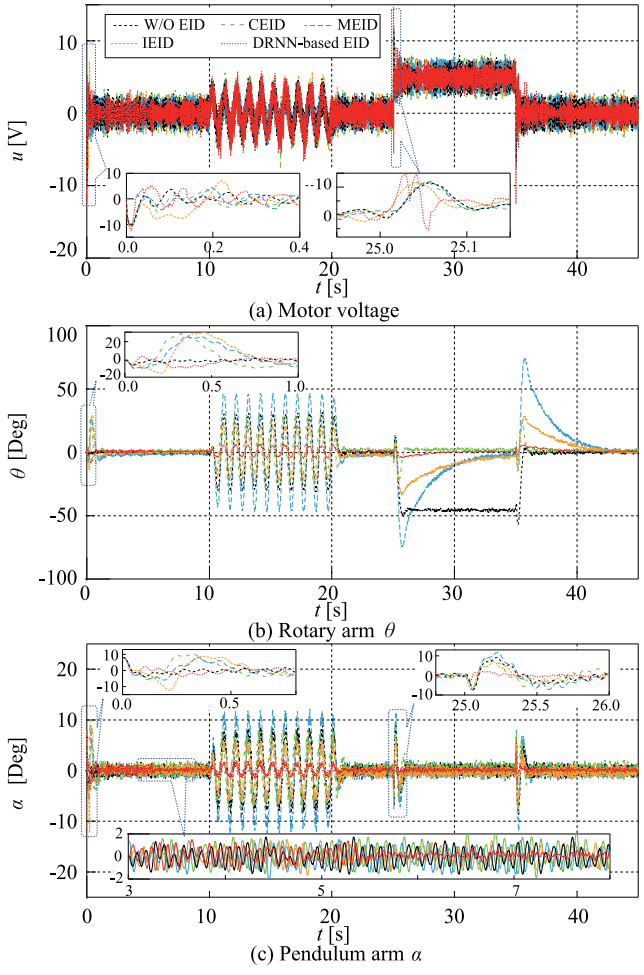


Fig. 5. Comparative results of control methods without the EID estimator, with the EID estimator, with the IEID estimator, the MEID estimator, and with the developed DRNN-based estimator.

rejection performance index was established to evaluate the impact of nonlinearities and disturbances on the system output. Then, an adaptive law was developed in the first rough learning control stage with exponential decay guarantee of the performance index. A gradient-descent algorithm was used to train the feedforward NN to refine the disturbance-rejection performance in the second refined learning control stage. Further, the closed-loop control performance was analyzed and designed with integrated disturbance-rejection performance. Finally, the developed DRNN-based control system was validated by rotary inverted pendulum experiment comparisons with other methods.

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